Project Cost and Effort Estimation Techniques

To make the **software development process** **profitable** a **manager** **need** to make a **fair** **estimation** of the **development cost** and **manpower** required. Several **techniques** are available for **estimation** of **cost** and **effort** required in development **using** **size** of the s/w product.

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Project Estimation techniques

- Project planning involves **estimation of** different parameters such as **effort** required to develop the software, project **duration**, and **cost**.
- **These estimates** not only help in quoting the project cost to the customer, but are also **useful in resource planning and scheduling**.
- There are three broad categories of estimation techniques:
  - **Empirical estimation techniques**
  - **Heuristic techniques**
  - **Analytical estimation techniques**

Empirical estimation techniques

- **Empirical estimation** techniques are based on **making an educated guess** of the project parameters **with help of prior experience with developing similar products**.
- Though **this technique is based on common sense**, different activities involved in **estimation have been formalized over the years**.
- Two popular empirical estimation techniques are:
  - **Expert Judgment Technique**
  - **Delphi Cost Estimation**

Expert Judgment Technique

- Expert judgment is one of the most **widely used estimation techniques**.
- In this approach, an **expert makes an educated guess** of the problem size after analyzing the problem thoroughly. Usually, the expert estimates the cost of the different components (i.e. modules or subsystems) of the system and then combines them to arrive at the overall estimate.
- **Demerits**:
  - This technique is **subject to human errors**, inadvertent **overlooking of some of the factors** and **individual bias**.
  - Further, an **expert** making an estimate **may not have experience and knowledge** of **all aspects** of a project.
- A **more refined form of expert judgment** is the estimation made **by group of experts**. Estimation by a group of experts minimizes factors such as individual oversight or bias.
  - However, **estimate by a group of expert can be erroneous** because of any **political reason** or **dominance by a senior member** in the group.

Delphi cost estimation

- Delphi estimation is carried out **by a team** comprising of a **group of experts** and **a coordinator**.
- The **coordinator provides each estimator** with **a copy** of the software requirements specification **(SRS) document** and a form for recording his cost estimate.
- **Estimators complete their individual estimates anonymously** and submit to the coordinator.
  - In their estimates, the estimators mention any unusual characteristic of the product which has influenced his estimation.
  - The **coordinator prepares and distributes the summary of the responses of all the estimators**, and includes any unusual rationale noted by any of the estimators.
  - **Based on this summary, the estimators re-estimate**.
  - **This process is iterated** for several rounds. However, **no discussion among the estimators is allowed** during the entire estimation process.
- **After several iterations** of estimations, the **coordinator compiles the results** and **preparing the final estimate**.

Heuristic Techniques

- **Heuristic techniques assume** that the **relationships among the different project parameters** can be **modeled** using **suitable mathematical expressions**.
- Once the **basic** (**independent**) parameters **are known**, the **other** (**dependent**) parameters **can be** easily **determined by** substituting the value of the basic parameters in **the mathematical expression**. Different heuristic estimation models can be divided into the following two classes:
  - **Single variable model**
  - **Multi variable model**

Heuristic Techniques Single variable estimation

- **Single variable estimation** models provide a means to estimate the desired characteristics of a problem, using some previously estimated basic (independent) characteristic of the software product such as its size. A single variable estimation model takes the following form:

$$\text{Estimated Parameter} = c_1 * e^{d1}$$

- **e is the characteristic of the software which has already been estimated (independent variable)**. **Estimated Parameter** is the **dependent** parameter to be estimated.
- The dependent parameter to be estimated could be effort, project duration, staff size, etc. **c1 and d1 are constants**. The values of the constants c1 and d1 are usually determined using data collected from past projects (historical data).
- The **basic COCOMO (COnstructive COst estimation MOdel)** is an example of **single variable cost estimation model**.

Heuristic Techniques multivariable variable estimation

- A multivariable variable estimation model takes the following form:

$$\textbf{Estimated Resource} = c_1 * e_1^{d1} + c_2 * e_2^{d2} \ldots$$

- Where e_1, e_2, … are the basic (independent) characteristics of the software already estimated, and c1, c2, d1, d2, … are constants.
- **Multivariable estimation models** are expected to **give more accurate estimates** compared to the single variable models, since a **project parameter is typically influenced by several independent parameters**.
- The independent parameters influence the dependent parameter to different extents. This is modeled by the constants c_1, c_2, d_1, d_2, … .
- Values of these constants are usually determined from historical data.
- The **intermediate COCOMO** model can be considered to be an **example of a multivariable estimation model**.

COCOMO COnstructive COst estimation MOdel (**Boehm, 1981**)

- Boehm postulated that any software development project can be classified into **three categories (Organic, Semidetached and Embedded)** based on the development.
- Roughly speaking, **application programs** such as data processing software are considered in **first category**, **utility programs** such as **Compilers**, **linkers**, etc are in **second category** and **system programs** such as **Operating systems** and real-time system programs, etc. in **third category**.
  - **System programs** interact directly with the hardware and typically involve meeting timing constraints and concurrent processing and accordingly **cost of development is quite high** whereas **application program development cost is low**.

Categories of software

- **Organic**:
  - A development project can be considered of organic type, if the project deals with developing a well understood application program, the size of the development team is reasonably small, and the team members are experienced in developing similar types of projects.
- **Semidetached**:
  - A development project can be considered of semidetached type, if the development consists of a mixture of experienced and inexperienced staff. Team members may have limited experience on related systems but may be unfamiliar with some aspects of the system being developed.
- **Embedded**:
  - A development project is considered to be of embedded type, if the software being developed is strongly coupled to complex hardware, or if the stringent regulations on the operational procedures exist.

COCOMO Model (Boehm, 1981)

- According to Boehm, s**oftware cost estimation should be done through three stages**:
  1. **Basic COCOMO**
  2. **Intermediate COCOMO**
  3. **Complete COCOMO**
- In **basic COCOMO model**, development **effort** and **time required** for software development **are estimated considering only product size**. These **estimates are further refined using intermediate and complete COCOMO**.

Basic COCOMO Model

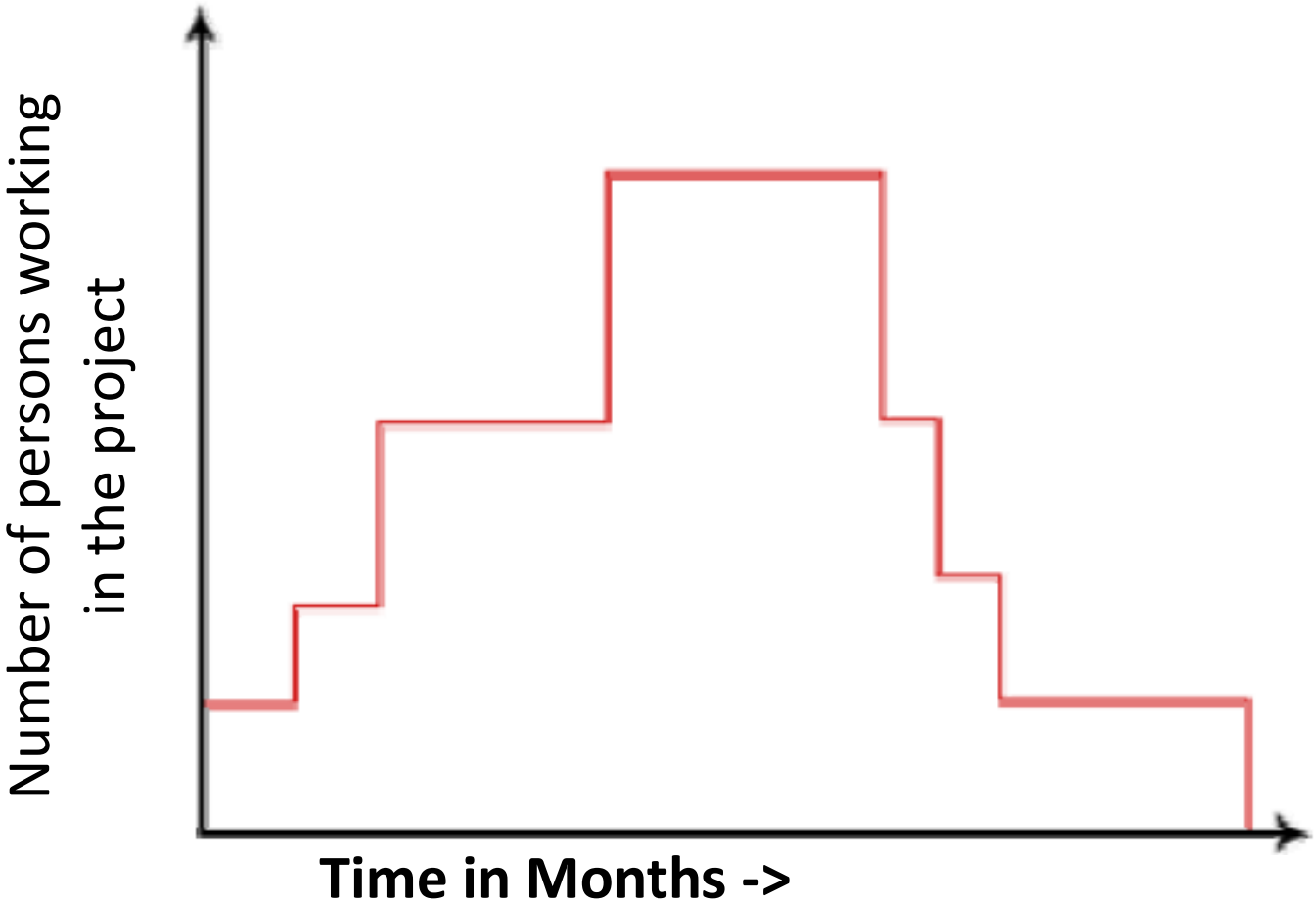
- The basic COCOMO model gives an approximate estimate of the project parameters. The basic COCOMO estimation model is given by the following expressions:

$$\textbf{Effort} = a_1 \times (\textbf{KLOC})^{a2}\ \textbf{PM}$$

$$\textbf{Tdev} = b_1 \times (\textbf{Effort})^{b2}\ \textbf{Months}$$

- Where
  - **KLOC** is the **estimated size** of the software product expressed in Kilo Lines of Code,
  - a_1, a_2, b_1, b_2 are constants those differ for each category of software products,
  - **Effort** is the total effort required to develop the software product, expressed in **person months (PMs)**.
  - **Tdev** is the **estimated time** to develop the software, expressed in months,

Basic COCOMO Model



- The effort estimate is the area under the person-month plot (as shown in fig). It should be carefully noted that an effort of 100 PM does not imply that 100 persons should work for 1 month nor does it imply that 1 person should be employed for 100 months.

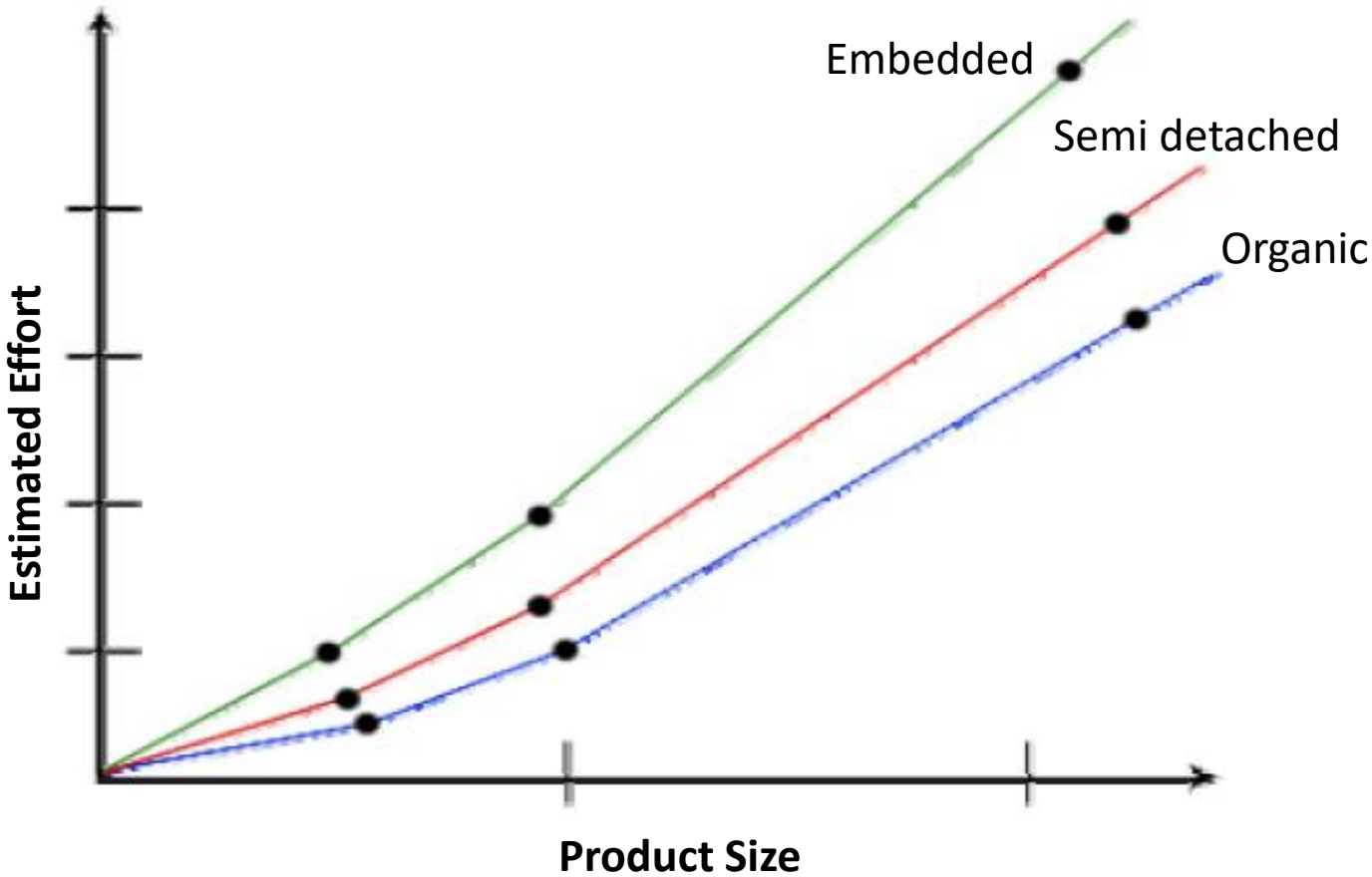
Basic COCOMO Model

- **Estimation of development effort**
  - For the three classes of software products, the formulas for estimating the effort based on the code size are shown below:
  - **Organic : Effort = $2.4(KLOC)^{1.05}$ PM**
  - **Semi-detached : Effort = $3.0(KLOC)^{1.12}$ PM**
  - **Embedded : Effort = $3.6(KLOC)^{1.20}$ PM**
- **Estimation of development time**
  - For the three classes of software products, the formulas for estimating the development time based on the effort are given below:
  - **Organic : Tdev = $2.5(Effort)^{0.38}$ Months**
  - **Semi-detached : Tdev = $2.5(Effort)^{0.35}$ Months**
  - **Embedded : Tdev = $2.5(Effort)^{0.32}$ Months**
- **Boehm derived the above expressions by examining historical data collected from a large number of actual projects**.

Basic COCOMO Model

- **Q.** Estimated **size** of a S/W is **500 KLOC .** What is the **cost** and **effort** required if the S/W is of types **organic**, **semi detached** and **embedded**?
- Estimation of development **effort**
  - **Organic: Effort = $2.4(500)^{1.05}$ PM= 1637.31 PM**
  - **Semi-detached: Effort = $3.0(500)^{1.12}$ PM = 3162.04 PM**
  - **Embedded: Effort = $3.6(500)^{1.20}$ PM = 6238.30 PM**
- Estimation of development **time**
  - **Organic : Tdev = $2.5(1637.31)^{0.38}$ = 33.33Months**
  - **SemiDetached:Tdev=$2.5(3162.04)^{0.35}$ = 41.967Months**
  - **Embedded : Tdev = $2.5(6238.30)^{0.32}$ = 40.960 Months**

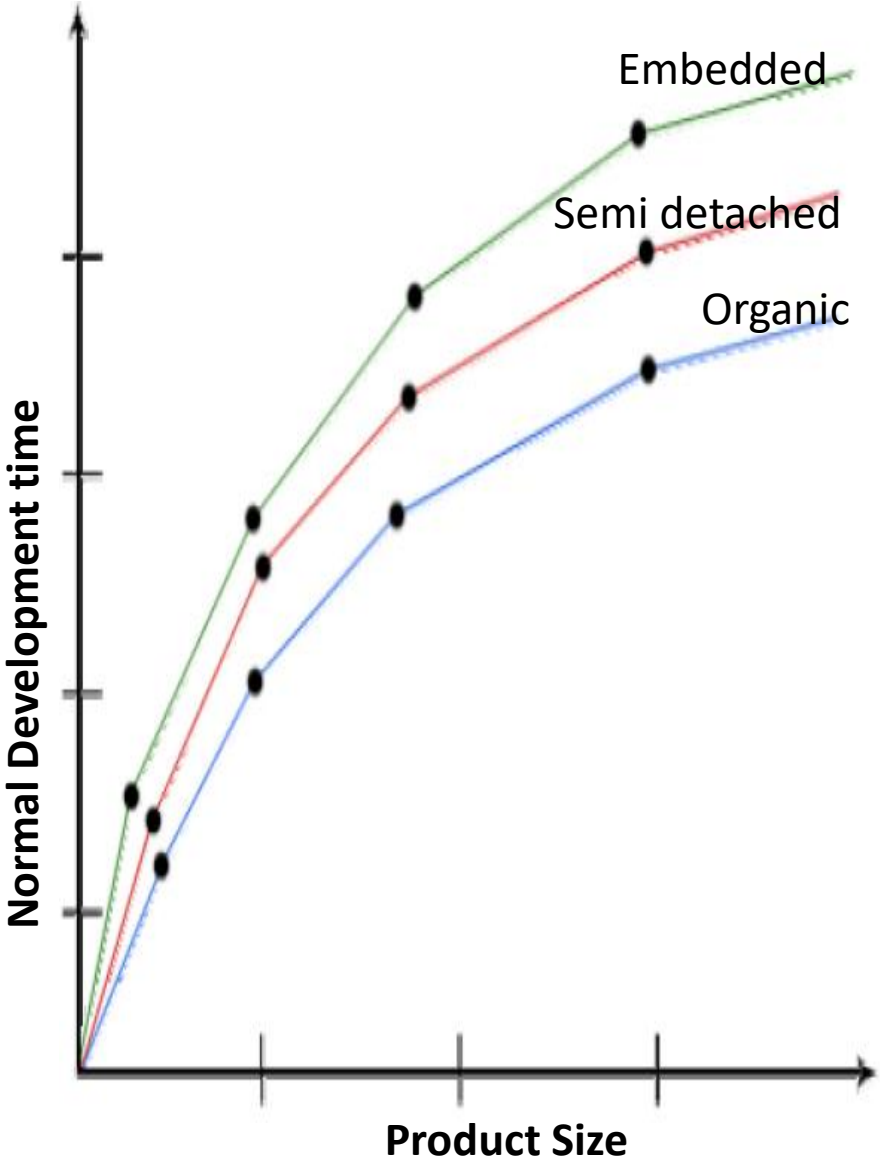
Basic COCOMO Model



we can observe that the effort is somewhat super-linear in the size of the software product. Thus, the effort required to develop a product increases very rapidly with project size.

Basic COCOMO Model

- The **development time is** a **sub-linear** function of the size of the product, i.e. when the size of the product increases by two times, the time to develop the product does not double but rises moderately.
- This is because
  - for larger products, a larger **number of activities** which **can be carried out concurrently** can be identified.
  - The **parallel activities** can be carried out simultaneously. This **reduces the time** to complete the project.



Basic COCOMO Model

- It is important to note that the effort and the duration **estimations obtained** using the COCOMO model **are called as nominal** **effort estimate and nominal duration estimate**.
- The term nominal implies that **if anyone tries to complete the project in a time shorter** than the estimated duration, then the **cost will increase drastically**.
- But, **if anyone completes the project over a longer period** of time than the estimated, then **there is almost no decrease in the estimated cost** value.

Intermediate COCOMO model

- The **basic COCOMO model assumes** that **effort and development time** are **functions of the product size alone**.
  - However, a lot of other project parameters affect the effort required to develop the product as well as the time to develop.
- The **effect of all relevant parameters must be taken into account** to obtain an accurate estimation of the effort and project duration.
- The **intermediate COCOMO model refines the initial estimate** obtained using the basic COCOMO expressions by **using a set of 15 cost drivers (multipliers)** based on various attributes of software development. For example:
  - **If modern programming practices are used**, the **initial estimates are scaled downward** by multiplication with a cost driver having a value <1.
  - **If there are stringent reliability requirements** on the software product, this **initial estimate is scaled upward** by multiplication with a cost driver having a value >1.

Intermediate COCOMO model

- The project manager **rate 15 different parameters** for a particular project on a scale of **one to three**.
- Then, depending on these ratings, appropriate cost driver values were suggested which should be multiplied with the initial estimate obtained using the basic COCOMO.
- In general, the cost drivers can be classified as being attributes of the following items:
  - **Product:** The **characteristics of the product** that are considered include the **inherent complexity of the product**, **reliability requirements of the product**, etc.
  - **Computer:** The **characteristics of the computer** that are considered include the **execution speed required**, **storage space required** etc.
  - **Personnel:** The **attributes of development personnel** that are considered include the **experience level** of personnel, **programming capability**, **analysis capability**, etc.
  - **Development Environment:** Development environment attributes capture the **development facilities available to the developers**. An important parameter that is considered is the sophistication of the **automation (CASE: Computer aided Software Engineering) tools used for software development**.

Complete COCOMO model

- A **major shortcoming** **of both the basic and intermediate COCOMO models** is that they **consider a software product as a single homogeneous entity**.
  - However, **most large systems** are **made up several smaller sub-systems**. These **subsystems may have** widely **different characteristics**.
- The complete COCOMO model considers these differences in characteristics of the subsystems and estimates the effort and development time as the sum of the estimates for the individual subsystems.
- The cost of each subsystem is estimated separately.
- This approach reduces the margin of error in the final estimate.

Complete COCOMO model Example

- A distributed Management Information System (MIS) product for an organization having offices at several places across the country can have the following sub-components:
  - **Database part**
  - **Graphical User Interface (GUI) part**
  - **Communication part**
- The **communication part** can be considered as **embedded**, the **database part** could be **semi-detached**, and the **GUI part organic**.
- The costs for these three components can be estimated separately, and summed up to give the overall cost of the system.
- **For present day software development the COCOMO model is further improved in to COCOMO-2** in order to estimate time and effort required in different phases of development life cycle.